BOUNDARY TECH GENERATION / AGES 12–14

Level 3 — Junior Developers

A complete eight-session pathway for ages 12–14: projects, teaching steps, evidence, tools and child-safety boundaries.

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What this level is designed to teach

Junior Developers is the proposed programme for 12–14 years, with 80–90 minutes guided sessions. Teaching priorities: Variables, tests, interfaces, sensor thresholds. By the end, the learner should be able to present a project such as exportable game; multi-screen app; sensor robot, and explain at least one problem they solved.

Eight-session progression and evidence

Week	Theme	Age-appropriate deliverable
1	Imagine and start	Create user story and design brief
2	Build the world/interface	Build reusable interface and scene layout
3	Control and response	Use variables/state or sensor logic
4	Rules and objectives	Write expected/actual test cases
5	Feedback and accessibility	Add alternatives for input and feedback
6	Find and fix defects	Run repeatable tests and fix issue
7	Prepare a version	Export on a compatible machine and document limits
8	Private showcase	Explain a failed test and a fix

Allowed tools and safety boundaries

Scratch or Godot starter, MIT App Inventor prototyping and sensor kits after mentor preflight. No real personal or live location data.

Finished project acceptance

- Runnability on the approved workshop device.
- An original child-made idea or improvement that the learner can explain.
- At least one recorded or explained test, including what was expected.
- One adjustment made because of feedback or a defect.
- Local safe project packaging and only opt-in guardian-approved private sharing.

Facilitator and guardian notes

Age is a default placement guide, not a test of worth or ability. The host and parent can choose accommodations, quiet presentation and slower pace. A workshop is not independent childcare, a credit-bearing qualification or a commercial data contribution. The original blueprint is a proposal: there are no confirmed dates, named registered students or verified screening outcomes to publish.